Ultra-SMR and Its Field Adoption

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Shingled Magnetic Recording (SMR) increases recording density, but early adoption was limited by software and system constraints. This paper reviews the concept of Ultra-SMR and discusses how zoned software support, solid state drive (SSD) tiering, cloud architectures, and emerging AI workloads have accelerated SMR deployment in high-capacity storage systems.

Index Terms— Perpendicular Magnetic Recording (PMR), Heat Assisted Magnetic Recording (HAMR), Conventional Magnetic Recording (CMR), Shingled Magnetic Recording (SMR).

I. Introduction

The concept of shingled magnetic recording (SMR) was introduced in the early 2000s [1],[2]. The initial motivation for SMR was to overcome the media / magnetic recording trilemma [4]. In the late 1990s it was recognized that there were some scaling limitations to magnetic recording that could be characterized as a three-way trade-off between (i) thermal stability (KuV/kT); (ii) writability (available write field); and (iii) signal/noise or grain size scaling [3].

SMR addresses the trilemma by keeping the write head dimensions (and write field) constant while continuing to scale the track width. Unlike in Conventional Magnetic Recording (CMR), tracks are written in an overlapping manner, like shingles on a roof. The non-overlapped portions of the tracks will contain the information. This process allows data to be written sequentially but removes the ability to update data in place. SMR architecture divides the Hard Disk Drive (HDD) into zones where each zone must be written to sequentially and can only be rewritten after the zone is reset, effectively erasing all the data in the zone. Zones can be written to independently and reads can be performed without constraints. New interface standards were introduced to accommodate this architecture: Zoned Block Commands (ZBC) in INCITS T10 [5] and Zoned Device ATA Command Set (ZAC) in INCITS T13 [6].

SMR architecture was introduced to address the trilemma, but energy assisted techniques were also introduced, such as Energy-Assisted Perpendicular Magnetic Recording (ePMR), Microwave-Assisted Magnetic Recording (MAMR) and Heat Assisted Magnetic Recording (HAMR). It is therefore natural to ask if SMR can be combined with these techniques. Studies have shown [7]-[9] that when combined with energy assisted techniques (e.g. ePMR, HAMR) SMR architecture continues to provide the highest recording density.

II. ultra-smr

SMR architecture provides intrinsic benefits that go beyond addressing the trilemma. It minimizes adjacent track interference, since a single adjacent track will be written exactly once. This also implies there is no double-sided squeeze. In addition, by controlling the direction of shingling, we can make sure we always have a clean edge between tracks [9].

Ultra-SMR is an extension of the original SMR architecture that takes advantage of the sequential write only nature of SMR to introduce further recording density gains that are independent of the underlying recording technology.

Error correcting code technologies such as data interleaving and track level coding are not suitable for CMR. Interleaving a logical sector into several physical sectors or adding parity sectors to a track require read-modify-write operations to update logical sectors, resulting in unacceptable performance impact. For SMR, there is virtually no performance penalty with significant recording density gains [10]. Ultra-SMR technologies are relatively new, and we can expect larger recording density gains in the future.

The result of the Ultra-SMR approach is significant recording density advantages that are independent of the underlying recording technology (PMR, ePMR, MAMR, HAMR, etc).

III. Integrating SMR into storage solutions

When SMR was first introduced, figuring out how to fit it into existing storage architectures appeared to be a daunting task. Several approaches were attempted ranging from introducing indirection systems in the HDD, like the flash translation layer in SSDs, to host-based translation layers. An extensive survey of the many publications studying such architectures can be found in [11].

Initial adoption of SMR was limited due to ill-suited traditional storage architectures and the significant software development required.

IV. Software ecosystem

The first practical software support for SMR was the libzbc open-source library, developed by Damien Le Moal, introduced in 2014 [12]. Libzbc was pivotal for SMR not only because it was the first software tool that allowed users to integrate SMR HDDs into their systems, but also because it provided an emulation mode that allowed software engineers who only had access to CMR HDDs to start developing solutions for SMR.

When SMR HDDs were first introduced in the marketplace, there was no support beyond libzbc. Initial rudimentary support was added to the Linux Kernel in 2015, and it wasn't until the early 2020s that general purpose filesystems were introduced that could natively support SMR. See Figure 1.

Today the situation is much different with mainstream filesystems such as XFS [13] having native support for SMR, enabling a distributed storage system to be setup to take advantage of the capacity gains and reliability improvements of SMR without the need to write a single line of code [14], [15].

In addition to the open-source software support, two developments help accelerate the adoption of SMR, namely the introduction of SSDs in storage architectures and cloud computing.

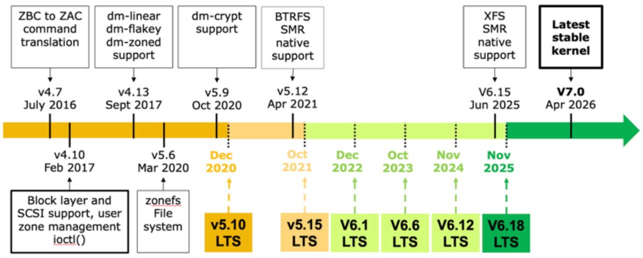

Fig. 1. Evolution of SMR support in the Linux Kernel.

V. Introduction of SSDs

With the introduction of SSDs into storage architectures, a shift in workloads and requirements for HDDs started to happen. As SSDs became more mature and enterprise capable, they started to be integrated into mainstream storage architectures. SSDs are typically used as part of a tiered solution where small and latency sensitive IOs and data that needs frequent updates are sent to the SSD tier. This means that the workload going to HDDs became cooler, more sequential and with significantly fewer updates. This substantially reduces performance differences in the overall system between using CMR and SMR in the HDD tier.

VI. The transition to cloud computing

Cloud computing has been a transformative force that completely changed the architecture of IT systems. In storage one of the early significant breakthroughs came with the Google File System (GFS) [16]. Since then, many distributed storage solutions have emerged with variations on the themes introduced in GFS. These solutions have in common the process of spreading data among several independent storage servers. In the early implementations, it was common to have three or more copies of the data stored in different servers to protect against failures. Over time, these solutions migrated to use erasure codes to reduce the amount of storage overhead used for reliability. The transition to erasure codes makes it naturally more expensive to perform updates in place since data and parity updates require read-modify-write operations involving multiple servers. As a result, systems migrated to an architecture where new data is always appended with a background process performing garbage collection as needed [17],[18]. The HDD tier in these systems sees a mostly append-only workload with periodic garbage collection, a workload for which there is very little difference in performance between CMR and SMR based HDDs.

VII. AI and the future of storage

Cloud architectures have taken over the industry during the past two decades. The transition to cloud architectures together with the evolution of the software ecosystem have propelled Ultra-SMR into the main architecture in use for HDDs today.

AI workloads are set to dominate growth for the next several years. AI workloads are more sequential and throughput bound, further playing to the strengths of Ultra-SMR. AI is generating an unprecedented amount of data and an unsatiable demand for storage. We expect Ultra-SMR architecture to continue to play a central role in the foreseeable future.

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